

# REPORT\_11

This report will talk about:

Stepper motors

Parameters of a DC motor

## **STEPPER MOTORS:**

Stepper motors are motors which are used as actuators to move robotic arms and for other tasks.

A stepper motor consists(usually) of a magnetised core(permanent magnet)(rotor) which is placed in a stator consisting of electromagnets.

The stator electromagnets are divided into groups of which the opposite magnets are in the same groups. Now the magnets are switched on group wise so that the rotor magnets align themselves according to the stator magnet orientation. Once the alignment is achieved, the group of active magnets is changed thus forcing the rotor magnets to realign themselves. This is done in a way such that the alignment motion sums up to give us rotation in one direction. Now the angle the motor covered in one step, is called the step angle. This depends a lot on the way the magnet orientation is changed ( different types of cycles: half stepping, microstepping etc).

The way the magnets are switched also affects the step resolution and

Now stepper motors have a high pole count, infact some stepper motors have a tooth count of 50, along with 8 electromagnets in the stator, each of those electromagnets further having their surface divide into teeth. This creates a very small step angle and makes transitions more precise.

Stepper motors have high torque as their speed is low, one of the main advantage a stepper motor has when it comes to being an actuator is that it doesn't use a feedback loop to resend its orientation again and again. Instead, it rotates with respect to number of steps corresponding to the input voltage, in this motor, the predefined quantity is the number of steps rather than which step the motor is at when a particular voltage is applied.

## **PARAMETERS OF A DC MOTOR:**

When reading a DC motor datasheet, there are certain quantities which come in very handy as some of these are quantities which specify the safe limits of the working of a motor.

Nominal Voltage: this is the voltage at which all the other quantities are measured ( also the suggested voltage as it is the most efficient)

No load speed: this is the speed of the motor when it does not have any load.

Stall torque: it is the maximal torque which can be applied before the motor comes to a rest.

Back emf constant: a constant which provides a direct relation between back emf and velocity.

Stall current: the current consumed by a motor when it is at rest( ot start).

Rated speed, torque and current: these provide the values of speed, torque and current at maximum efficiency.

Maximum speed, current, torque: the limit values of the motor ( changing from continuous to intermittent operation regions).

Intermittent and continuous regions are two regions of operations for a motor.

In the continuous region, the motor can be used for a long time as the load on the motor is not enough to damage it if used for a long time.

The intermittent region is the region which if used for a long time will damage the motor.

The load in this region is enough to make the motor heat up and lead to damage.

Other parameters usually include information such as its dimensions and structures, direction of rotation of a motor when it is connected in a circuit, weight of the motor, motor armature winding resistance and inductance, back emf and torque constants, moment of inertia etc.